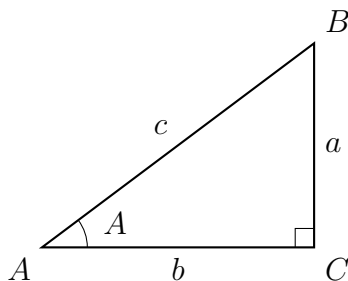


Find and Fix: Legs, Hypotenuses, and the Pythagorean Theorem

Directions

- The theorem is $a^2 + b^2 = c^2$, where c is always the **hypotenuse** — the side opposite the right angle, and the longest side.
- To find the hypotenuse, **add** the squares of the legs. To find a leg, **subtract** the known leg's square from the hypotenuse's square. Decide which one you need before you write anything.
- Where a student's work is shown, correct it *and* say what went wrong.
- Round any answer that is not exact to two decimal places.



How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. A right triangle has legs of 9 cm and 12 cm. Find the length of the hypotenuse.

Answer: _____ cm

2. A right triangle has a hypotenuse of 25 cm and one leg of 7 cm. Find the length of the other leg.

Answer: _____ cm

3. Jamal was asked for the missing side of a right triangle with a hypotenuse of 13 cm and one leg of 5 cm. He wrote:

$$\sqrt{13^2 + 5^2} = 13.93 \text{ cm}$$

(a) Work out the correct length.

Answer: _____ cm

(b) Describe what Jamal did wrong.

4. Find the distance between the points $(-2, 1)$ and $(4, 9)$, in grid units.

Answer: _____

5. A ladder 17 ft long leans against a vertical wall with its foot 8 ft from the base of the wall. Priya was asked how far up the wall the ladder reaches. She wrote:

$$\sqrt{17^2 + 8^2} = 18.79 \text{ ft}$$

(a) Work out the correct height.

Answer: _____ ft

(b) Describe what Priya did wrong.

6. A right triangle has a hypotenuse of 15 and one leg of 9. Setting the missing leg to x gives the equation $x^2 + 81 = 225$.

Both solutions of the equation: $x =$ _____

Explain which of the two solutions can be the length of the leg, and why the other one cannot.

7. A rectangular garden measures 24 m along one side and 10 m along the other. A straight path runs from one corner to the opposite corner. How long is the path?

Answer: _____ m

8. A right triangle has one leg of 20 cm and a hypotenuse of 29 cm. Two students found the second leg. Marisa wrote $\sqrt{29^2 + 20^2} = 35.23$ cm. Ben wrote $\sqrt{29^2 - 20^2} = 21$ cm.

(a) The correct length of the second leg:

Answer: _____ cm

(b) Explain how you could tell which answer had to be wrong without checking the arithmetic.
